

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (IT) Examination

Dec. - Jan. 2014

IT702 High Performance Computer Networking

Date: 01.01.2014, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as directed

- (a) WI-FI Router works at layer -3. Justify. [01]
- (b) What is NNI in the context of ATM network? [01]
- (c) Define SSID. [01]
- (d) In ATM network Virtual Path is a subset of Virtual Circuit Identifier. [TRUE/FALSE] Justify. [01]
- (e) Define Network Congestion. [01]
- (f) Compare Circuit switching and Packet switching. [05]

- Q - 2
- (a) Explain the call control procedure for frame relay. [03]
 - (b) What is the effect of congestion on packet switching networks? [02]
 - (c) Compare Static routing vs. Dynamic routing. [04]
 - (d) Show the comparison of ATM and Frame relay protocol stack. [06]

OR

- Q - 2
- (a) Compare ABR vs. GFR ATM network attributes. [03]
 - (b) What is the significance of Cell loss priority (CLP) field of ATM protocol header? [02]
 - (c) Explain congestion control in packet switching networks. [04]
 - (d) Explain the use of token bucket scheme in ISA approach. [06]
- Q - 3
- (a) Describe wireless sensor network applications. [05]
 - (b) Explain hardware constraints associated with the wireless sensor network. [05]

OR

- Q - 3
- (a) Explain characteristics of real time transport protocol. [05]
 - (b) Explain MPLS in brief. [05]

SECTION – II

Q - 4 Do as directed

- (a) Does the GFC field of ATM protocol stack exist in UNI? [01]

[P.T.O]

- (b) Define Elastic Traffic. [01]
- (c) X.25 is a circuit switching network. [TRUE/FALSE].Justify. [01]
- (d) ATM Cell consists of _____ bytes of header. [01]
i) 3 ii) 6 iii) 4 iv) 5
- (e) What is forward explicit congestion notification (FECN)? [01]
- (f) Compare and contrast the ISA and DSA services. [05]
- Q - 5 (a) Discuss Choke packet congestion control technique. [03]
- (b) AAL 5 is also known as simple and efficient adaptation layer (SEAL). Justify. [02]
- (c) Explain Backpressure congestion control technique. [04]
- (d) Explain Integrated Service Architecture implemented in router. [06]

OR

- Q - 5 (a) Discuss various categories of explicit congestion signaling approach. [03]
- (b) What is committed information rate (CIR) in the context of frame relay congestion control? [02]
- (c) Compare distance vector routing vs. link state routing. [04]
- (d) Compare the different queuing schemes used in ISA approach. [06]
- Q - 6 (a) What do you mean by real time and non real time services with respect to ATM network? Explain ATM service category. [05]
- (b) Explain ATM protocol architecture. [05]

OR

- Q - 6 (a) Explain TP, VP and VCI in the context of ATM networks. [05]
- (b) List and explain the ISA approach and Functions in detail. [05]
